HealthConnect Rwanda: Week 8 Final Model and Data Science Package

> **Track:** Data Science
> **Client:** HealthConnect Rwanda
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> **Programme:** AnalystLab Africa Experience Lab
> **Project phase:** Week 8 - Final Integration, Documentation and Presentation

Project Overview

All records in this project are synthetic data created by the AnalystLab Africa community for the internship. This notebook continues the Week 6 and Week 7 work. It summarizes the Week 7 model tests, incorporates relevant Data Analytics findings, checks the admission-target lineage in the supplied workbook, and records the final integration decision.

The project workbook `HC_Clinical_Transactions.xlsx` contains an `Admissions` worksheet. The Week 6 master creates `admission_indicator` by marking an appointment positive when its ID matches an ID in that sheet. The source audit finds 586 matching admission rows and 4,414 admission rows whose appointment IDs do not match the appointment table. The admission source exists in the supplied project files, but the unmatched keys cannot be reconciled from the available columns and are documented as a dataset limitation.

Notebook Structure

Part I - Week 7 to Week 8 Transition

1. Summarize the Week 7 results, refinements and remaining issues
2. Include Data Analytics findings that affect feature and business decisions

Part II - Week 8 Final Integration and Model Readiness

1. Audit the target against source records and explain the unmatched keys
2. Set prediction-time and feature requirements
3. Confirm the final model decision and give evidence-based recommendations for weak performance
4. Document business suitability, limitations, cross-track contribution and ML Engineering requirements
5. Provide a concise model summary and presentation-ready findings

This notebook documents an analytical prototype and a readiness decision. It does not authorize clinical use or claim no-show prediction capability.

Part I - Week 7 to Week 8 Transition

The Week 8 brief asks for a short transition from the previous work, without repeating the full Week 7 report.

Transition item	Week 7 to Week 8 summary
Main Week 7 output	Tested admission-classification workflow comparing a prior-probability reference, Week 6-style Logistic Regression, refined Logistic Regression and Random Forest
Main test result	On the Nov-Dec holdout, refined Logistic Regression AP was 0.124 and ROC-AUC 0.466; prior reference AP was 0.134 and ROC-AUC 0.500
Major issue	The candidates did not beat the simple reference. The label is generated from linked Admissions rows, but 4,414 source keys are unmatched and may affect label completeness
Refinements	Removed the direct target copy and constant feature, retained appointment rows, added strictly prior appointment history, used chronological splits and compared two model types
Component ready	The model testing and evidence package is ready for review. No predictive model is approved for operational use
Still to integrate	The 4,414 unmatched keys cannot be reconciled from the supplied files; retain the label as a linked-record proxy and do not treat unmatched rows as confirmed non-admissions
Tracks involved	Data Analytics for the supplied validation findings; Data Science for the label, features and evaluation; ML Engineering with Charles for ongoing handoff alignment; Project Management for final readiness
Week 8 demonstration	Show the Week 7 comparison, the source-key audit, the no-go decision and the evidence needed before model selection can resume

Interpretation: Week 7 did not demonstrate model improvement. The supplied Admissions worksheet was used to construct the label. Since most admission keys do not match appointments, the label remains a proxy and the model errors are not confirmed clinical outcomes.

Week 7 Results Retained for Traceability

These scores came from the Week 7 notebook and use the label then available in the cleaned master. They describe performance against the linked-record proxy, not a clinically validated admission outcome; it is a proxy constructed from linked records in the supplied files.

Candidate	Average precision	ROC-AUC	Precision at 0.50	Recall at 0.50	F1 at 0.50	Balanced accuracy
Prior-probability reference	0.134	0.500	0.000	0.000	0.000	0.500
Week 6-style Logistic Regression	0.131	0.467	0.120	0.473	0.192	0.469
Refined Logistic Regression	0.124	0.466	0.120	0.445	0.190	0.471
Refined Random Forest	0.138	0.493	0.103	0.055	0.071	0.491

Average precision (AP) is the primary ranking metric for the imbalanced positive class; ROC-AUC is complementary, while precision, recall, F1 and balanced accuracy depend on the stated 0.50 threshold. The paired bootstrap interval for refined Logistic Regression's AP difference versus the prior reference was -0.023 to 0.017 and included zero. No meaningful improvement was demonstrated. The 4,414 unmatched admission keys are an additional reason to qualify the Week 7 metrics.

Data Analytics Findings Relevant to Data Science

The validation document provides these cross-track findings. They are attributed to Data Analytics; this notebook does not independently recalculate every dashboard test.

Finding	Data Science implication
KPI reconciliation and date-slicer checks passed	Supports the reported dashboard measures, not the admission-model target
Matched-bill amounts were similar to unmatched bills (KS p=0.66); status and severity shares differed by under one percentage point	Billing findings can be generalized cautiously with the matched-subset caveat; billing is not added to this admission model
Matched monthly billing trend slope was -0.3M/month (p=0.37), with 5.9% variation	No sustained billing trend was demonstrated; this is not an attendance trend
Facility rating tests p=0.79/0.80 and department rating p=0.29 were not significant	Exclude rating measures. `departmentname` is a category, not a rating
Revenue Summary 1.13bn did not reconcile to 9.31bn billing and was removed from the dashboard	Do not use this summary as a model feature or reconciled KPI
No attendance/no-show field was found	No missed-appointment model can be trained from the supplied outcome fields
Gender/chronic condition considered usable; age, prior appointment count, visit type, department, month, weekday usable with caveats; diagnosis/severity held for timing review	Carry these as provisional feature decisions and verify availability at prediction time. Prior-admission history needs key validation

Feedback linkage was 1,756 of 21,064 responses (about 8%). Linked/unlinked complaint rates were 40.9%/40.4%; mean ratings 2.93/2.99 (p=0.05). This may support a broader project insight with the linkage caveat, but feedback is not a predictor here.

Part II - Week 8 Final Integration and Model Readiness

2.1 - Admission Target and Source-Key Audit

The `Admissions` worksheet is present and was used to construct `admission_indicator`. The audit below reproduces the join and quantifies unmatched keys. A complete binary field is not enough by itself: source relationships also need to be credible.

```
Source-lineage audit:
                                                Check Observed
                                    Appointment rows     5000
                              Admissions worksheet rows     5000
                              Unique AdmissionID values     5000
                 Distinct Admission AppointmentID values     5000
                  Duplicate Admission AppointmentID rows        0
                Admissions rows linked to an appointment      586
            Admissions rows with unmatched AppointmentID     4414
                     Distinct linked appointment IDs      586
                    Positive labels in cleaned master      586
Master label equals recreated link flag for every row     True
```

Interpretation of the source-key audit. The repository workbook contains 5,000 Admissions rows. Of these, 586 `AppointmentID` values match the 5,000 appointment records and 4,414 do not. The cleaned master has 586 positive labels and reproduces the Week 6 rule: a matching admission row sets the appointment label to 1.

This confirms that the Admission worksheet exists and explains how the label was constructed. It does not explain why most source IDs are unmatched. Those records could reflect a key-format, source-scope or referential-integrity problem; no automatic ID repair should be applied. The 0 labels currently mean “no linked Admissions row found,” not necessarily a confirmed clinical non-admission. The separate finding that no attendance/no-show field is available remains applicable.

2.2 - Prediction Time and Feature Decisions

The proposed prediction point is **after the initial clinical assessment and before the admission decision/outcome**, once each selected field has actually been recorded. This is a project assumption: the supplied files do not establish the recording time of every clinical field. A booking-time model cannot use a diagnosis or severity value recorded later; an arrival-time model needs its own timestamped feature set.

Feature	Week 8 decision	Reason
Gender, chronic condition	Potentially usable	Data Analytics identified these as potentially usable; inspect their definitions and missingness in the supplied fields
Age, prior appointment count, visit type, department, month and weekday	Potentially usable with caveats	Check availability at the stated prediction time and define the history window from the supplied dates
Diagnosis and severity	Hold	The supplied data do not establish whether these fields precede the admission decision; exclude them unless reliable timestamps demonstrate this
Prior admission count/rate	Hold	The Admissions sheet exists, but 4,414 unmatched source keys may make this history incomplete
Facility/department ratings	Exclude	Data Analytics found no significant differences; department name itself is a separate category
Billing, payments, revenue, complaint/feedback	Exclude from this model	Different outcome and potentially observed after prediction; revenue summary failed reconciliation
`admissions_record_count`	Prohibit as a predictor	It duplicates the existing label and leaks it directly
Attendance/no-show outcome	Unavailable	The supplied dataset contains no attendance/no-show outcome; this project cannot evaluate attendance prediction

The Week 7 feature set is not approved unchanged. The unmatched admission keys and target semantics remain limitations of the supplied data. Retain only variables shown to be available before the outcome, then rerun temporal validation if the target or data are corrected.

2.3 - Final Candidate Selection

No predictive model is approved as the final operational candidate. Week 7 Logistic Regression did not beat the prior reference on average precision or ROC-AUC. The Random Forest showed severe overfitting, with training AP/ROC-AUC of 0.815/0.969 versus validation values of 0.117/0.499. These results provide no evidence that deploying the current model would improve decisions.

The target source does exist in the repository, but 4,414 of 5,000 Admissions keys do not match an appointment. The supplied files do not explain why these IDs are unmatched. A zero label means only that no Admissions row was linked; it does not confirm that no admission occurred. The Week 8 Data Science outcome is a documented **no-go decision for predictive use**, a traceable comparison, and an evidence-based plan to address weak performance and data quality.

2.4 - Final Integration Readiness and HC-POD Record

Readiness item	Current response	Status
Final component	Week 7 model-test evidence plus Week 8 source-key and readiness review	Evidence package ready to present

Readiness item	Current response	Status
Problem addressed	Whether current data support reliable admission prediction for the broader appointment-support goal	No-show objective remains unsupported
Current model status	No final candidate approved; Week 7 metrics use a linked-record proxy	Not ready for predictive use
Week 7 improvement	Leakage checks, chronological evaluation, candidate comparison and error analysis	Completed; no gain demonstrated
Source-key limitation	The supplied files contain 4,414 unmatched Admission IDs; their relationship to appointments cannot be determined from the available keys and columns	Unresolved in supplied data
Output for ML Engineering	If a model later passes validation: versioned feature schema, prediction time, fitted pipeline, input/output contract and validation summary	No model artifact to deploy now
Readiness evidence	Week 7 outputs, executed source-key audit, Analytics validation summary and issue register	Review package ready; model integration not ready
Remaining limitations	4,414 unmatched admission keys, no no-show label, diagnosis/severity timing, one-year history, weak model ranking	Open

Cross-Track Integration Record

Required item	Week 8 record
Track collaborated with	Data Analytics (Agu Precious); ML Engineering (Charles; reviewed the work and shared the handoff checklist; technical alignment is ongoing as of 2026-11-07); Project Management for final readiness
Output received	Data Analytics validation summary with dashboard, billing, rating and target-inventory findings; Charles's ML Engineering checklist for a future handoff
Output to provide	Week 7 model results and this audit: 5,000 Admissions rows, 586 matching keys, 4,414 unmatched keys, and the Week 6 label rule. The repository and results were shared with Charles; he replied with ML Engineering handoff requirements, and technical alignment is ongoing as of 2026-11-07. GitHub: https://github.com/Dorsin/healthconnect-rwanda
Final integration activity	Data Analytics findings were incorporated and the join was audited. The unresolved keys and proxy target are documented; retesting is conditional on consistent target links and timestamped features being available in the supplied data. Charles reviewed the work and shared requirements for input specifications, target definition, preprocessing, model compatibility, outputs, validation and reproducibility. Technical alignment is ongoing; no candidate is ready for integration
What changed	The local Admissions worksheet is present, but 4,414 keys remain unmatched in the supplied data. The Week 7 candidate remains provisional and prior-admission features are held. Charles's response added an ML Engineering handoff checklist; technical alignment is now in progress
How this improves the solution	Prevents unsupported target assumptions from being treated as confirmed outcomes and aligns future Data Science outputs with ML Engineering requirements
Evidence	Data Analytics validation summary, executed audit in Section 2.1, and Charles's reply with ML Engineering handoff requirements
Retest	Not performed: the available files do not provide a mapping for the unmatched keys. Repeat temporal evaluation only if the supplied target data are corrected or regenerated with consistent keys

The Data Analytics findings and Charles's ML Engineering handoff checklist are documented. Technical alignment is ongoing. The key mismatch and retest remain dataset limitations; no model artifact is ready for integration.

2.5 - Information to Provide to ML Engineering

The repository and Week 7/Week 8 model summary have been shared with Charles in the ML Engineering track. He reviewed the work and replied with a handoff checklist covering input specifications, target definition, preprocessing, model requirements, outputs, validation evidence and reproducibility. Technical alignment is ongoing. The handoff remains a **no-deployment notice and future requirements checklist**. There is no model artifact to deploy because the tested candidates have not shown useful performance and source-key quality needs review. Repository: https://github.com/Dorsin/healthconnect-rwanda

Deployment information	Current status / requirement
Target definition	Week 6 used `admission_indicator = 1` when an Admissions `AppointmentID` matched an appointment; the resulting label is a linked-record proxy, and unmatched rows are not confirmed negatives in the supplied files
Prediction point	Proposed after initial clinical assessment and before the admission decision. The supplied files do not establish when every feature is recorded, so use only fields shown to be available by this point
Final model/artifact	None approved. Do not integrate the Week 7 candidate as a decision service
Feature schema	Provisional; include only fields present and available before prediction in the supplied data. Exclude admission history, diagnosis and severity because their links or timing are not established
Input/output behavior	If a future model is approved, document required types, missing values, unseen categories, probability meaning, decision threshold and error behavior
Dependencies/configuration	Pin package versions, record configuration, source-data version, model version and reproducible build steps
Testing and monitoring	Add schema validation, unit/integration tests, logging, drift/performance monitoring and rollback instructions before any operational use
Use restrictions	No no-show prediction, clinical recommendation or automated scheduling decision is supported by current evidence

When a valid model exists, Data Science should send the serialized pipeline, exact input schema, preprocessing, output definition, threshold rationale, validation results by time/segment, known risks and version details. ML Engineering should not compensate for weak performance by masking it in the pipeline.

2.6 - Business Suitability and Non-Technical Model Summary

What the Week 7 results say: on the 821-row final holdout, the admission-link label prevalence was 13.4%. The prior reference had AP 0.134 and ROC-AUC 0.500. The selected refined Logistic Regression had AP 0.124 and ROC-AUC 0.466. At threshold 0.50, its proxy-label confusion matrix was 353 true negatives, 358 false positives, 61 false negatives and 49 true positives. Only 49 of 407 positive alerts were correct (about 12% precision), while recall was 44.5%. These counts are errors against the current linked-record proxy; they are not confirmed errors against clinical admission outcomes because 4,414 source IDs remain unmatched in the supplied files.

Strengths: Week 7 used a time-ordered holdout, compared candidates with a simple baseline, kept preprocessing inside model pipelines, checked a direct target copy and documented uncertainty. These are useful evaluation practices.

Weaknesses: The selected candidate ranked below the prior reference; the Random Forest overfit strongly; the proxy target has 4,414 unmatched admission keys; diagnosis/severity timing remains unconfirmed; and no attendance/no-show label exists. Current metrics do not support operational use.

Business decision: do not use the current model scores to prioritize patients, change care, or automate appointment actions. Treat the scores as an educational analysis output only; this synthetic dataset and its current results do not

support operational or clinical decisions. No-show prediction requires a separately defined and validated attendance outcome.

2.7 - Recommendations for Weak Model Performance

The current results call for a controlled improvement plan. Moving to more complex algorithms or lowering the threshold will not fix an unreliable label or create predictive signal.

1. **Pause operational use.** The refined model AP (0.124) is below test prevalence (0.134) and ROC-AUC is 0.466. Random Forest AP (0.138) is only slightly above prevalence, ROC-AUC is 0.493, and recall is 5.5% at 0.50. Neither candidate shows reliable ranking.
2. **Investigate the target links.** Inspect the 4,414 unmatched `AppointmentID` values using only the supplied files: compare data types and formatting, whitespace and leading zeros, table grain, duplicates and workbook scope. The available files provide no authoritative mapping for these rows. Do not fuzzy-match them or treat them as confirmed negatives; document the mismatch as unresolved.
3. **Align target and business objective.** Admission prediction and missed-appointment prediction are different tasks. If attendance is the goal, the supplied dataset must contain an attendance/no-show outcome before attendance prediction can be evaluated.
4. **Confirm prediction time and feature availability.** Exclude diagnosis/severity until timestamps prove they precede the decision. Hold prior-admission features until history links are validated. Exclude post-outcome billing, feedback and admission counts.
5. **Rebuild simple benchmarks after validation.** Start with a prevalence reference and regularized Logistic Regression. Add more models only after target and feature quality are sound. Fit imputation, encoding, scaling, class weights or resampling on training data only.
6. **Use future-oriented validation.** Keep a final chronological holdout untouched; with more years, add rolling-origin validation. Choose thresholds on validation data against agreed alert capacity and error costs. Report average precision/PR curve, ROC-AUC, precision, recall, F1, calibration and confusion counts, with uncertainty intervals and segment checks where sample sizes permit.
7. **Set a stakeholder acceptance rule.** Agree on minimum improvement over the reference, acceptable alert volume, recall and calibration. If no model meets the rule, record no-go and request better labels/features instead of deployment.
8. **Retest changes on fresh evidence.** If a corrected version of the supplied dataset becomes available, rerun the same temporal evaluation and compare on an untouched holdout. Record source version, target logic, feature list and before/after results.

Model Findings for the Final Presentation

- **Contribution:** tested a time-aware admission-classification workflow and reviewed its errors.
- **Evidence:** refined Logistic Regression AP 0.124/AUC 0.466 versus prior-reference AP 0.134/AUC 0.500; Random Forest showed severe overfitting.
- **Decision:** no candidate is approved for operational use; the results show no reliable improvement.
- **Data issue:** 586 of 5,000 Admissions keys link to appointments; 4,414 remain unmatched in the supplied files; their meaning cannot be established from the available fields.
- **Business limit:** the available target is admission linkage, not missed-appointment attendance; no-show prediction is unsupported.
- **Next step:** use a target with consistent links and features available before the decision, then rebuild and retest before any ML Engineering integration.

Week 8 Task and Output Coverage

Week 8 task/output	Notebook evidence	Status
1. Review Week 7 testing	Part I transition and results	Complete; label caveat included
2. Confirm final candidate	Section 2.3	No candidate approved
3. Compare against baseline	Week 7 results table	Complete; same test rows, provisional labels
4. Present relevant metrics	AP, ROC-AUC, recall, prevalence and bootstrap interval	Complete
5. Explain strengths and weaknesses	Sections 2.3, 2.6 and 2.7	Complete
6. Document modeling decisions	Feature decisions and no-go decision	Complete
7. Explain business suitability	Section 2.6	Not suitable for operational use
8. Incorporate Data Analytics findings	Part I analytics summary and source-sheet clarification	Complete
9. Communicate requirements to ML Engineering	Section 2.5	Repository and no-deployment summary shared; Charles replied with the ML Engineering requirements checklist; technical alignment ongoing as of 2026-11-07
10. Document limitations and risks	Sections 2.1-2.6	Complete; key mismatch documented as unresolved in the supplied dataset
11. State permitted and prohibited uses	Section 2.6	No clinical, automated scheduling or no-show use
12. Prepare model content for presentation	Section 2.7	Slide-ready model findings included
13. Prepare non-technical model summary	Section 2.6	Complete
14. Organize a professional notebook/report	Numbered sections, interpretations and evidence tables	Complete
Final candidate model	Sections 2.3 and 2.5	None approved or deployable
Evaluation and baseline comparison	Week 7 results table	Included; based on linked-record proxy
Error analysis summary	Section 2.6 confusion counts	Included; errors are proxy-label errors and require caution
Feature and model decisions	Sections 2.2-2.5	Included
Cross-track contribution	Section 2.4	Analytics findings recorded; Charles provided ML Engineering handoff requirements; technical alignment ongoing; key mismatch and retest documented as dataset limitations
Model documentation and presentation material	Sections 2.5 and 2.7	Included

Week 8 Data Science Project Summary

Required summary item	Current record
Final Data Science component	Week 7 model-test evidence and Week 8 target-link/readiness decision
Week 7 result	Refined Logistic Regression AP 0.124/AUC 0.466; prior reference AP 0.134/AUC 0.500 on the linked-record label
Most important new finding	The local Admissions worksheet has 5,000 rows; 586 keys match appointments and 4,414 do not. The source exists, but the unmatched keys cannot be reconciled from the files provided
Final candidate	None approved because results are weak and target-link quality is unresolved
Cross-track work	Data Analytics validation summary received; Charles reviewed the work and provided ML Engineering handoff requirements; technical alignment ongoing; key mismatch and retest documented as dataset limitations
Business relevance	Prevents unreliable scores from entering a workflow; no-show prediction remains unsupported
Remaining limitations	Unmatched admission keys, target semantics, diagnosis/severity timing, one-year history, no attendance label
Required next action	Maintain the no-go decision with the current files; a retest is only relevant if the supplied dataset has consistent target links and pre-decision feature timing
Readiness decision	Evidence package ready for presentation; predictive model and ML Engineering integration are not ready

This package summarizes Week 7 without repeating its full report. It records the source worksheet that exists, the unresolved key-quality issue, and a defensible no-go decision for the current models.